



Graph-based adaptive feature fusion neural network model for person-job fit

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Abstract

Online recruitment services are rapidly transforming traditional hiring practices in the job market. Accurate person-job fit is crucial for intelligent recruitment. Previous studies on person-job fit fail to explore job seekers' resume information from a multi-perspective approach, and neglect the sustainable learning of resume features. To address this, the present paper proposes a Graph-based Person-Job Fit Neural Network Fusion (GPJFNNF) model. Specifically, the model first generates local semantic representations of job requirements and resume text using the BERT model. Next, a graph structure is constructed based on historical successful recruitment records, and the constructed resume graph is input into a graph neural network to obtain a global semantic representation of the resume. Finally, the adaptive feature fusion mechanism is used to fuse the local and global semantics of the resume, and the final semantic representation of the resume, along with the semantic representation of the job requirements, being input into the person-job fit layer. Experimental results demonstrate that the proposed model achieves 94.63%, 94.15%, 95.04%, and 94.59% in the person-job fit task in terms of accuracy, precision, recall, and F1, respectively, significantly outperforming state-of-the-art baselines.

Keywords Person-job fit · Talent recruitment · Graph neural network · Adaptive feature fusion

Introduction

Traditional offline recruitment methods heavily rely on manual resume screening, resulting in time-consuming processes with issues of low efficiency and high costs. With the rapid development of the internet, vast amounts of job postings and resume information from job seekers have been generated, making online recruitment emerge as the mainstream mode of hiring. Consequently, the challenge for modern businesses lies in effectively selecting suitable talent from the multitude of resumes submitted by job seekers.

Person-job fit, referred to matching jobs with suitable job seekers, has been of great interest to many researchers. As a critical task for job recruitment, person-job fit task has been

well studied from different perspectives, such as big five personality factor model [1], job recommendations [2–4], etc. However, these efforts often rely on manual inspection by domain experts of resume and job key features, resulting in issues of low accuracy, high costs, and low recruitment efficiency.

In recent years, several researchers from the area of machine learning and natural language processing have also started to focus on person-job fit task. With the resurgence of deep learning techniques, person-job fit tasks has been well studied from different perspectives [5–9]. Researchers have designed various deep neural networks approaches to handle person-job fit, such as CNN-based neural networks [5], RNN-based neural networks [6], and bert-based neural networks [8, 9]. Among these researches, this task is viewed as a supervised text matching problem, aiming at measuring the matching degree between job positions and resumes. For example, Zhu et al. [5] proposed a job-resume matching modeling approach based on Convolutional Neural Networks (CNN). This method, known as PJFNN (Person-Job Fit Neural Network), employs two parallel neural networks to represent job and resume texts. In [8], Lavi et al. used fine-tuning Siamese SentenceBERT model to Match Jobs and

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Job Seekers. However, previous studies have analyzed the person-job fit task from a single perspective, failing to leverage multi-perspective feature extraction from job seekers' resumes. This limitation hinders the comprehensive capture of the complex and dynamic demands of person-job fit. Furthermore, previous researches overlook the continuous learning of job seekers' resume features during the training process, making it difficult for models to retain and utilize key features in resumes over the long term.

To address this, this paper proposes a novel Graph-based Person-Job Fit Neural Network Fusion (GPJFNNF) model. Inspired by the success of graph neural networks in the field of text matching [10–12], graph neural networks are introduced into the task of person-job fit. Inspired by the successful application of the KAN network [13] in fusing features and fitting complex functions, KAN network is introduced into the person-job fit task. Specifically, first, the BERT model is used to generate the local semantics of job descriptions and applicants' resume texts. Second, based on historical successful recruitment records, a resume graph structure is constructed for the applicants, which is then fed into the graph neural network, incorporating an attention mechanism to further obtain a global semantic representation of the applicants' resume texts. Then, the KAN network (Kolmogorov–Arnold Networks) is applied to fuse the dual-perspective features of local and global semantics of the applicants' resumes, and the semantic similarity is used to calculate the matching degree between the final semantics of the job descriptions and the applicants' resumes. Finally, several state-of-the-art baselines are used to evaluate the GPJFNNF model on recruitment data. The experimental results validate the effectiveness of the GPJFNNF model.

The main contributions of this paper are as follows:

- (1) The dual-perspective features are used to deeply mine job seekers' resume information, and historical recruitment data is introduced to enhance the model's ability to capture dynamic changes.
- (2) To the best of our knowledge, this is the first time that the KAN network has been introduced into the person-job fit task. An adaptive feature fusion mechanism is designed to enhance the model's continuous learning ability and mitigate forgetting the key features of job seekers' resumes during the learning process, further enhance the learning of the matching between job requirements and the potential capabilities of applicants.
- (3) A novel Graph-based Person-Job Fit Neural Network Fusion (GPJFNNF) model has been proposed, and experiments have shown that the proposed GPJFNNF model provides optimal performance.

The rest of the paper is structured as follows. In “Related works” Section, some related work of recruitment analysis and text mining with deep learning are discussed. In “Methods” Section, the technical details of our Graph-based Person-Job Fit Neural Network Fusion (GPJFNNF) model will be introduced. In “Experiments” Section illustrates recruitment dataset, experimental setup, baseline methods, and evaluation metrics, and then comprehensively evaluate the performance of the proposed approach. Finally, in “Conclusion” Section, the conclusion of the paper has been made.

Related works

In this section, some works related to this paper will briefly be introduced, the related works can be grouped into two categories, namely recruitment research and text mining with deep learning.

Recruitment researches

The mismatch between employees and jobs positions can lead to employee turnover behavior [14]. Organizations need to spend time and cost recruiting suitable talents. Recruitment research is the key to supporting the success of organizations, being a crucial aspect of talent management. More and more researches on recruitment analysis have garnered significant attention from researchers. Harris used quantitative methods to analyze hiring analysis, analyzing the differences in job requirements between technical recruitment and nontechnical recruitment [15]. There also exist some researches focused on the career development of talents. Li et al. provided career path development advice from a survival analysis perspective [16]. Wang et al. proposed a good time of making a job transition [17]. Some researchers are devoted to studying recruitment market from a novel perspective [18], using talent flow analysis [19] and market trend analysis [20].

Recruitment has also been studied from other angles, such as performance prediction [21–23], information extraction from resumes [24], etc. In recent years, research on person-job fit problem, specifically the matching of job recruitment information with resume information, has garnered significant attention from researchers. In this research area, some researchers have drawn inspiration from recommendation systems to study the task of person-job fit, namely job-resume matching. Malinowski and other researchers [25], employing a manual extraction approach, extracted features from job seekers' resumes and company job postings. They designed a system for person-position recommendations to facilitate the matching of candidates with positions. Lee et al. [2] proposed a job recommendation algorithm based on the preferences of job seekers and job requirements. Gopalan et al. [4] designed a content-based recommendation algorithm that

matches job requirements and resume information. Rácz and others [26] used machine learning methods to explore hierarchical relationships between recruitment text and resume text, achieving personalized resume recommendations.

Early researches treat person-job fit task as a recommendation problem, but recent research studies person-job fit problem from another perspective. The literature [5] introduced a job-resume matching modeling approach based on Convolutional Neural Networks (CNN), aiming to match suitable individuals to appropriate job positions. This method, known as PJFNN (Person-Job Fit Neural Network), employed a parallel neural networks (such as a job neural network and a resume neural network) to semantically represent job and resume texts. Semantic similarity is calculated to measure the match between job positions and job seekers.

Along this line, the literature [6] achieved better results via employing BiLSTM and attention mechanism instead of CNN. Qin et al. proposed an ability-aware person-job fit method, first used BiLSTM to encode the word-level representations of job seekers' resumes and job requirements, and then employs hierarchical attention mechanisms to further capture the different importance of the skills (job experience) in job seekers' resumes and job requirements. Le et al. [7] presented an effective and interpretable person-job fit approach, first encoded the resume and job requirements using CNN and a collaborative attention mechanism, and then computed the intention scores of job seekers and recruiters through the bilateral intention networks.

In the literature [8], Lavi et al. designed a Siamese SBERT model by fine-tuning way, called conSultantBERT, which focusing on constructing textual embeddings in vacancies and resumes by resume-vacancy pairs labeled. Shao et al. [9] have proposed a job-resume matching method that explored the internal and external interactions of multiple attributes between the job post and the resume. Primarily, BERT was used to project the keys, values, and corresponding attribute sources of each attribute in the job post and the resume into the same semantic space. After that, the key embeddings and the source embeddings were progressively fused into the value embeddings. Eventually, the aggregation matching layer was employed to predict the matching degree.

However, previous studies fail to consider multi-perspective feature extraction from job seekers' resumes. Previous researches overlook the continuous learning of job seekers' resumes features during the training process.

Text mining with deep learning

In recent years, the field of text mining has attracted widespread concern from the researchers in this academic field, such as, sentiment analysis [27], software defect detection [28], online harassment detection [29], phishing email identifica-

tion [30], and examining public perceptions of virtual voice assistants [31].

In general, research on job-resume fit based on texts is classified as text mining tasks. Due to the good performance of deep neural network, more and more researchers tend to use deep neural network to tackle text mining tasks. DSSM [32] is a deep neural network (DNN) model for semantic similarity via leveraging DNN to represent each input sentence.

Among various deep neural network models, convolutional neural network (CNN) and recurrent neural network (RNN) are two representative and widely-used deep neural network architectures in text mining. As one of the representative works in this area, dynamic convolutional neural network (DCNN) is proposed to model sentences in text classification tasks [33]. He et al. [34] employed CNN to extract semantic features from multiple levels of granularity. The literature [35] leveraged multi-column convolutional neural networks to deal with question answering tasks.

Compared to CNN, RNN-based models are more suitable for tackling sequential texts. RNN-based approaches have attracted much more attentions on text mining tasks, such as sentiment classification [36], semantic similarity assessment of sentence pairs [37], personality trait recognition [38], etc. For instance, tang et al. [36] leveraged gated recurrent neural network to handle document level sentiment classification. Zhang et al. [39] presented a novel deep recurrent neural network model for automatically extracting keyphrases from tweets. Xue et al. [38] employs BiGRU to extract the left and right context semantics of words which aimed at precisely recognizing multi-labeled personality traits from texts.

Other neural network methods are also popular in text mining tasks, such as graph neural network and BERT [40]. Recently, inspired by the success of graph neural network, some researchers started to focus on using graph-based neural network methods to handle text mining tasks. For example, zhang et al. [41] used a graph convolutional network to learn the representations, and modeled the local interactions via attention mechanism based for addressing the short-long text matching problem. Furthermore, Sentence-BERT (SBERT) [42] used siamese and triplet network structures to represent semantically sentence embeddings, which reaches remarkable performance.

Methods

This section presents the technical details of the Graph-based Person-Job Fit Neural Network Fusion (GPJFNNF) model. As illustrated in Fig. 1, the GPJFNNF model consists of three main components: the BERT-based local semantic representation of job requirements and resumes, the graph network-based semantic representation of resumes, and the person-job fit interaction.

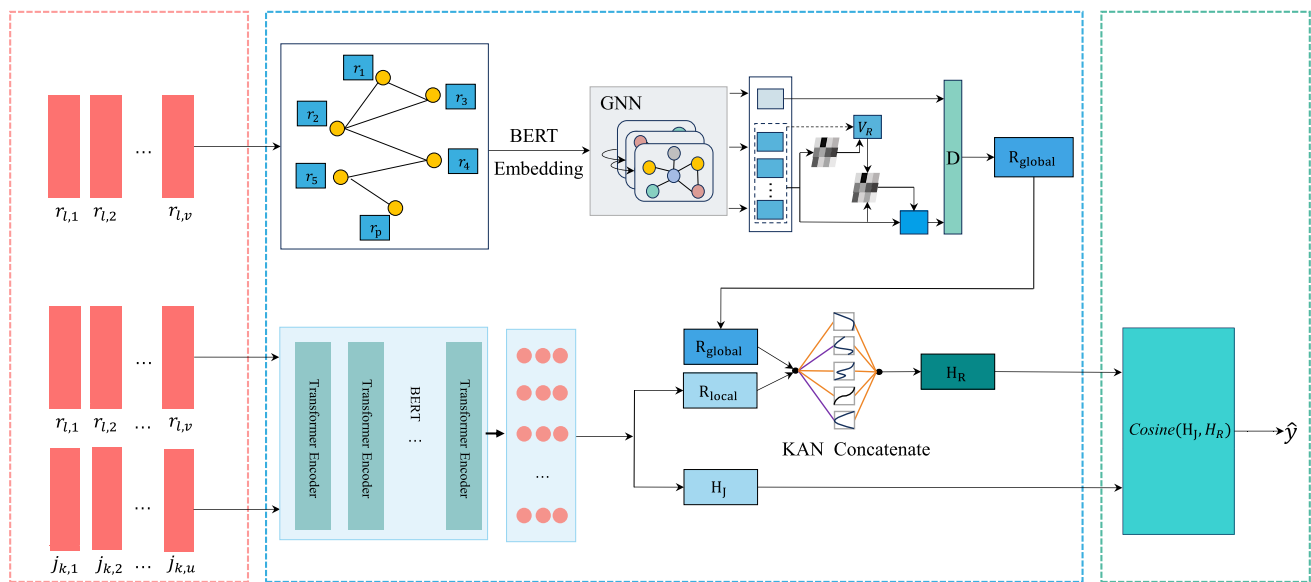


Fig. 1 Architecture of the proposed GPJFNNF model $j_{k,1} \dots j_{k,u}$ and $r_{l,1} \dots r_{l,v}$ represent the job requirements and resume experience descriptions, respectively, and they are input into the neural network. R_{global} denotes the global semantic representation of the resume. H_J and R_{local}

correspond to the local semantic representations of the job requirements and resume. H_R represents the final resume semantic representation obtained by the KAN network fusion of R_{global} and R_{local} . $\hat{y}$ indicates the person-job fit result

Specifically, job requirements and resume experience descriptions are fed into the neural network as input, and the output is the recruitment result label (1 indicating a successful hire, 0 indicating a failed hire). In the BERT-based local semantic representation component, the BERT model is used to model personal resumes and company job postings, generating local semantic representations for both. For the graph network-based resume semantic representation module, a graph structure is constructed based on historical successful recruitment records, and the constructed resume graph is input into a graph neural network. An attention mechanism is then introduced to identify the importance of different skills in the job requirements and resume experiences, further obtaining the global semantic representation of the resume. Finally, the KAN network is used to fuse the local and global semantics of the resume, and the final semantic representation of the resume and the job requirement representation are fed into the person-job fit interaction layer, automatically determining the matching result between the resume and the job posting.

Problem formulation

This paper focusing on the matching between job requirements and the experiences listed in applicants' resumes aims to study the person-job fit task. The job-seeking process consists of a job requirement, an applicant's resume, and the job application record. Let J represent a job, which includes o job requirements, denoted by $\{j_1, j_2, \dots, j_o\}$.

Each job requirement j_k consists of u words, represented as $\{j_{k,1}, j_{k,2}, \dots, j_{k,u}\}$. The same job J may have multiple different job applications. Similarly, let R represent an applicant's resume, which contains p experiences (such as work experience, project experience, or competition experience), denoted by $\{r_1, r_2, \dots, r_p\}$. Each experience r_l consists of v words, represented as $\{r_{l,1}, r_{l,2}, \dots, r_{l,v}\}$. S_h represents the historical successful recruitment records, with S denoting the set of all job-seeking processes, where each s_i is composed of a pair $\{(J, R)\}$. s_i is denoted as $s_i = (j_i, r_i, y_i)$, where the label y_i indicates the record of the job application. $y_i = 1$ signifies a successful hire, while $y_i = 0$ indicates a failed application. The objective of this paper is to learn a person-job fit prediction model that computes the similarity for each pair $\{(J, R)\}$ to predict the outcome of future job applications.

BERT-based local semantic representation of job requirements and resumes

To effectively capture the sentence embedding semantic information of job requirements and resume texts, this paper introduces the BERT model [40] to extract the local semantics from job requirements and resume texts.

Specifically, a pre-trained BERT model is used to perform sentence embeddings on the input resume information and job requirements, generating corresponding semantic vectors to obtain local semantic representations that include multiple relationships. First, the resume information and job require-

ment sequences are passed through the word embedding and position embedding layers. Next, linear transformations layers are applied to separately obtain the query, key, and value vectors for the job requirements and resumes. By concatenating multiple attention heads, the model captures features from different sentence spaces within the job requirements and resumes, resulting in semantic representations that encapsulate the multi-relational aspects of job requirements and resumes. The representations are then subjected to Add & Normalize and Feed Forward Neural Network operations.

By inputting the words from each job requirement and resume text, the local semantic information containing multiple relationships between the job requirements and resume information is ultimately obtained. The semantic representation vectors for the job requirements and resumes, denoted as H_{local}^J and H_{local}^R , are calculated using the following formulas:

$$H_{local}^J = Bert(x_{j_{k,1}:j_{k,u}}) \quad (1)$$

$$H_{local}^R = Bert(x_{r_{l,1}:r_{l,v}}) \quad (2)$$

Here, $x_{j_{k,1}:j_{k,u}}$ represents the input sequence of job requirements $\{j_{k,1}, j_{k,2}, \dots, j_{k,u}\}$, and $x_{r_{l,1}:r_{l,v}}$ represents the experience sequence in the input resume text $\{r_{l,1}, r_{l,2}, \dots, r_{l,v}\}$.

Graph network-based resume semantic representation

To predict more accurately the outcome of a job applicant's future job-seeking process, it is essential to extract additional global information from historical successful recruitment records using a graph neural network (GNN). This information is used to further refine the BERT-based local semantic extraction of resumes after that.

In this section, the applicant's resume text is modeled using a graph neural network. The following subsections provide a detailed explanation of the resume graph construction method and the graph neural network-based resume representation.

Resume graph construction method

To capture the global semantic information of resumes from historical successful recruitment records using a graph neural network, firstly an undirected graph, named Graph R-R (denoted as G_{R-R}), represented by (V_R, E_R) , needs to be constructed. In this graph G , each node represents a resume, and the initial features of the nodes are embedded using the BERT model to enhance node representations, ensuring that

each node's initial features contain deep semantic information. The set of nodes V_R represents the current resume R_c to be predicted and the historically successful resumes related to R_c . The set E_R denotes the edges between these resumes.

Specifically, for each resume $R_i (R_i \in R_r)$, the goal is to predict whether it matches with a job requirement J_h . First, a set of resumes R_K is searched, satisfying the condition $(J_h, R_j, 1) \in S_h$, which indicates that this set contains all resumes related to R_i . Next, for these resumes, we construct the graph's adjacency matrix and calculate the edge weights in the matrix. For each R_i , the resumes $[R_1, R_2, \dots, R_p] \in R_r$ are treated as different individual nodes. A separate graph and its adjacency matrix are created for each resume R_i . The weights w of the edges are generated using a random number generation method. After obtaining the weights w for all edges, the adjacency matrix A_R is constructed.

Resume representation based on graph neural networks

After constructing the graph matrix, an adjacency matrix related to resume information is obtained, with each resume in Graph R-R being considered as an individual node. The representation of each resume node is learned through the update function of the GNN unit [43, 44], as shown in formula:

$$a_i^t = A_i \left([g_0^{t-1}, \dots, g_n^{t-1}]^T H^t + b^t \right) \quad (3)$$

$$z_i^t = \sigma \left(W_z^t a_i^t + V_z^t g_i^{t-1} \right) \quad (4)$$

$$r_i^t = \sigma \left(W_r^t a_i^t + V_r^t g_i^{t-1} \right) \quad (5)$$

$$\tilde{g}_i^t = \tanh \left(W_h^t a_i^t + V_h^t (r_i^t \odot g_i^{t-1}) \right) \quad (6)$$

$$g_i^t = (1 - z_i^t) \odot g_i^{t-1} + z_i^t \odot \tilde{g}_i^t \quad (7)$$

where $[g_0^{t-1}, \dots, g_n^{t-1}]$ represents the node vectors of resume information, A_i denotes the i row of the adjacency matrix corresponding to node i , and $H^t, W_z^t, V_z^t, W_r^t, V_r^t, W_h^t, V_h^t$, and b^t are learnable parameters. z^t and r^t refer to the reset gates and update gates, respectively, and $\odot$ denotes element-wise multiplication. σ represents the sigmoid function.

In addition, a trainable resume word embedding matrix H_{GR} is constructed. Semantic information from the resumes is embedded using this word embedding matrix to obtain g_0^R . The resume embedding matrix is then fed into the graph neural network to obtain the representations of all nodes in Graph R-R, denoted as g_R .

To further extract higher-level semantic representations of the resumes (i.e., global semantics of the resumes), the vector representations of all nodes in Graph R-R are mapped into the vector space of job requirements using a soft attention mechanism. The calculation formula is as follows:

$$V_J = \sum \alpha_i g_i^R \quad (8)$$

The weight coefficient α_i can be expressed as:

$$\alpha_i = q_r^T \sigma(W_r g_i^R + b) \quad (9)$$

where g_i^R represents the vector of each node in Graph R–R, q_r and W_r are learnable parameters, and V_J denotes the semantic representation of historically successful resumes in the job requirement vector space. V_J is used to assess the significance of experiences (e.g., skills) valued in historical hiring processes. The attention mechanism is employed to estimate the degree of match between each resume and V_J . The resume representations g_0^R obtained from the trained embedding layer in Graph R–R are concatenated with the experience representations e^R derived from interactions with job requirements and applicant information to obtain the global information of the current resume, denoted as H_{global}^R .

$$H_{global}^R = \tanh(W_R[g_0^R; e^R] + b_R) \quad (10)$$

where g_0^R represents the input resume representation, e^R denotes the experience relationship between the applicant's resume and the job requirements, and W_R and b_R are learnable parameters.

Person-job fit

The local semantic information of job requirements, H_{local}^J , is obtained by using a BERT-based job semantic representation method. For the resume text, both the local semantic information, H_{local}^R , and the global semantic information, H_{global}^R , are obtained by employing BERT-based semantic representation and graph network-based resume representation methods respectively. For job requirements, only the local semantics are used to represent the job posting text. Recently, inspired by some methods from the field of physics [13, 45], especially, the successful application of the KAN network in fusing features and fitting complex functions, KAN network is introduced into the person-job fit. For the resume text, the local and global semantics are concatenated, and a KAN network is used to integrate these semantics to represent the final semantic meaning of the resume text.

$$H_J = [H_{local}^J] \quad (11)$$

$$H_R = KAN([H_{local}^R, H_{global}^R]) \quad (12)$$

The adaptive feature fusion mechanism is utilized to learn activation functions, further enhancing the continuous learning of job seekers' resume texts, and effectively fusing both global and local semantics of the resumes. Each layer of the KAN network maps the input $[H_{local}^R, H_{global}^R]$ through a set

of learnable functions to enhance the integration of local and global semantics in the resume text, thereby improving the network's representation capability and adaptive ability.

After obtaining the semantic representations for both job requirements and resume text, cosine similarity is used to compute the person-job fit results. The calculation of cosine similarity is as follows:

$$c = \text{cosine}(H_J, H_R) \quad (13)$$

Experiments

In this section, the recruitment dataset, detailed experimental settings, baseline methods, and evaluation metrics will be introduced. Then, the performance of our proposed model and all baseline methods, followed by the elaborate analysis of the experimental results have been conducted.

Dataset

Experiments on a real-world recruitment dataset sourced from a company in China have been done. To protect the privacy of job seekers, all the recruitment records were anonymized by removing user profiles.

The experimental dataset includes information on recruitment job positions, job seekers' resumes, and the matching information between job positions and resumes. The recruited positions encompass roles such as software engineers, hardware engineers, micro-controller development engineers, and technical service engineers, among others. A single position can receive resumes from multiple job seekers, and conversely, a job seeker can submit resumes to multiple positions. Each job position contains an average of 3.95 job requirements. Each job seeker's resume includes an average of 2.82 work/project/competition experiences. Each job position contains an average of 77.15 words. Each job seeker's resume includes an average of 76.05 words. The statistical information of the recruitment dataset is presented in Table 1.

This dataset includes job position information, resume information, and label for recruitment results. The first job position column represents the recruitment requirements for the job position, the second resume describes the experiences in the job seeker's resume (such as work experience, project experience, or competition experience), and the third label indicates the label for the recruitment result. The person-job fit task fundamentally belongs to a binary classification problem. If the recruitment is successful, the label is 1; if unsuccessful, the label is 0.

Table 1 Statistical information of the dataset

Statistics	Values
Average job requirements per job posting	3.95
Average work/project/competition experiences per resume	2.82
Average words per job posting	77.15
Average words per resume	76.05
Average number of resumes per job posting history	29.7
Average number of job applications per candidate history	2.22
Positive instances	2126
Negative instances	2152

Experimental setup

In this subsection, a detailed explanation of the experimental setup is provided. All experiments in this work were conducted in Python, utilizing open-source deep learning frameworks such as PyTorch and tensorflow. The computer employed was comprised of an Intel(R) Core(TM) i7-13700KF CPU running at 3.40 GHz and a NVIDIA GeForce RTX 4060Ti GPU.

The recruitment data set was divided into a training set, validation set, and test set. 80% of the samples were used for training, 10% for validation, and 10% for testing. Node embeddings in the graph were initialized using the pretrained BERT-Base-Chinese language model, with an embedding dimension of 768. The number of GNN layers was set to 1, with a hidden layer size of 200. The number of KAN layers was set to 1. The max words length was set to 256. The learning of our model was conducted with a mini-batch size of 16 samples. The model was trained for 5 epochs. The AdamW optimizer was employed to train the model, aiming at optimizing the proposed GPJFNNF model. The learning rate is set to $2e-5$. The tuning experiments of the above hyperparameters are done on the validation set.

Baseline methods

To demonstrate the performance of the proposed GPJFNNF model, implementation is compared with some state-of-the-art baselines. Some traditional machine learning methods are employed, such as Decision Tree (DT), Support Vector Machine (SVM), Adaboost, Random Forests (RF), and Gradient Boosting Decision Tree (GBDT), XGBoost, and CatBoost as the baselines. For fairness, state-of-the-art job-resume matching methods and texts matching methods based on deep learning are selected as the baselines, which include DSSM, Siamese-LSTM, ABCNN, PJFNN, BPJFNN, APJFNN, SBERT, IPJF, conSultantBERT, and

InEXIT. Specifically, each method can be described as follows:

- **Classic traditional machine learning methods:** For the traditional machine learning methods, seven classic methods, including DT, SVM, Adaboost, RF, GBDT, XGBoost, and CatBoost are implemented. These methods are typical and representative. The representation is constructed using the Doc2Vec vectors as the input features of all baseline methods in job recruitment texts and resume texts.
- **DSSM:** DSSM [32] adopts the deep neural network (DNN) to learn dense feature representation. For person-job fit, DSSM is exploited to represent job posting texts and resume texts respectively, and computes the semantic similarity of the job and resume pair.
- **Siamese-LSTM:** Siamese Long Short-term Memory (Siamese-LSTM) [37] utilizes the same LSTM to encode sentences into sentence vectors by forward and backward directions in job posting and resume texts, and finally match a pair of job and resume.
- **ABCNN:** Attention Based Convolutional Neural Network (ABCNN) [46] is proposed to model a pair of sentences. Sentence pairs of job posting texts and resume texts are viewed as input, and predicted whether the recruitment is successful.
- **PJFNN:** PJFNN [5] is a CNN-based method for job-resume matching. It utilizes a parallel convolutional neural networks to effectively learn the joint representation of job postings and resumes. It takes a job-resume pair as input, and uses the cosine similarity to calculate semantic similarity of job posting and resume, ultimately predicting whether a candidate matches a recruitment position.
- **BPJFNN:** Basic Person-Job Fit Neural Network (BPJFNN) [6] is a simplified version of APJFNN, being an RNN-based model. It utilizes BiLSTMs to learn the semantic representation of each word in the job requirements and the experiences of job seekers. And then it uses an aggregation layer to predict the matching degree of job postings and resume of job seekers.
- **APJFNN:** APJFNN [6] employs BiLSTM to encode word-level features of resumes and job requirements. Subsequently, hierarchical attention mechanisms are introduced to extract the critical skills and work experiences from resumes and job requirements descriptions.
- **SBERT:** Sentence-BERT (SBERT) [42] is a modification of the pretrained BERT network. It utilizes Siamese BERT-Networks to derive semantically sentence embeddings. The job posting texts and the resume texts are considered as input of sentence pairs. The semantic sim-

ilarity between the job postings texts and the resume texts of job seekers is calculated by cosine-similarity.

- **IPJF**: An efficient and interpretable person-job fit network have utilized CNN and a collaborative attention mechanism to encode features of resumes and job requirements descriptions and computed intention scores between candidates and jobs by the bilateral intention networks [7].
- **conSultantBERT**: ConSultantBERT [8] is a fine-tuned siamese sentence-BERT (SBERT) model being used to match jobs and job seekers. The job posting texts and the resume texts pairs are considered as input and the model fine-tuned is based on siamese sentence-BERT.
- **InEXIT**: InEXIT [9] utilizes the internal and external interactions to concentrate on comprehensively modeling the semi-structured multivariate attributes of Job-resume matching.
- **GPJFNNF**: GPJFNNF (Graph-based Person-Job Fit Neural Network Fusion) model is the model proposed in this paper.

Evaluation metrics

The person-job fit task in this paper can be treated as a binary classification problem in fact, where the matching result is categorized as either positive or negative. Therefore, four evaluation metrics are employed to assess the performance of the GPJFNNF model.

These four evaluation metrics are accuracy, precision, recall, and F1-score (F1). On the test dataset, the model's performance is assessed using accuracy, precision, recall, and F1. Accuracy refers to the proportion of correctly predicted positive (successful recruitment matches) and negative (failed recruitment matches) samples out of all samples, i.e., the ratio of correctly predicted samples to the total number of samples. Precision represents the ratio of correctly predicted successful recruitment samples to the total number of samples predicted as successful recruitment, indicating the probability of correct predictions among all samples predicted as successful recruitment. Recall is the ratio of correctly predicted successful recruitment samples to the total number of actual successful recruitment samples. However, precision and recall are one-sided evaluation metrics. F1, determined by both precision and recall, is the harmonic mean of precision and recall, serving as a comprehensive evaluation metric. The indexes, including accuracy, precision, recall, and F1 evaluation metrics in the experiments, are all defined by True Positive (TP), False Negative (FN), False Positive (FP), and True Negative (TN). The mathematical formulas for calculating accuracy, precision, recall, and

F1 are as follows:

$$accuracy = \frac{TP + TN}{TP + FN + FP + TN} \quad (14)$$

$$precision = \frac{TP}{TP + FP} \quad (15)$$

$$recall = \frac{TP}{TP + FN} \quad (16)$$

$$F1 = \frac{2 \times precision \times recall}{precision + recall} \quad (17)$$

Next, the specific meanings of TP, FN, FP, and TN in this context is explained. True Positives (TP) are the samples that are both actual successful recruitments and correctly predicted as such. False Negatives (FN) are the samples that are actual successful recruitments but predicted as unsuccessful recruitments. False Positives (FP) are the samples that are actual unsuccessful recruitments but predicted as successful recruitments. True Negatives (TN) are the samples that are both actual unsuccessful recruitments and correctly predicted as such.

The results of the four evaluation metrics, including accuracy, precision, recall, and F1, are in the range of [0, 1]. Typically, a metric value of 0.5 indicates random guessing. A higher metric value indicates better classification performance of the model. For fairness, the same evaluation metrics are used for all methods in experiments.

Overall performance

The GPJFNNF model is compared with all baseline methods to evaluate the effectiveness of the proposed GPJFNNF model in person-job fit tasks. The compared methods in this paper include Decision Tree (DT), Support Vector Machine (SVM), Adaboost, Random Forests (RF), Gradient Boosting Decision Tree (GBDT), XGBoost, CatBoost, DSSM, Siamese-LSTM, ABCNN, PJFNN, BPJFNN, APJFNN, SBERT, IPJF, conSultantBERT, and InEXIT. Table 2 presents the experimental results for accuracy, precision, recall, and F1 metrics. The first column of the table lists the thirteen compared methods along with the proposed GPJFNNF model. The best results for each evaluation metric are highlighted in bold.

As shown in Table 2, the following observations can be made. In person-job fit tasks, the GPJFNNF model achieves experimental results of 94.63% accuracy, 94.15% precision, 95.04% recall, and 94.59% F1. Compared to DT, SVM, AdaBoost, RF, GBDT, DSSM, Siamese-LSTM, ABCNN, PJFNN, BPJFNN, SBERT, conSultantBERT, and InEXIT, the proposed GPJFNNF model outperforms all baselines in terms of accuracy, precision, recall, and F1. The experimental results demonstrate that the proposed GPJFNNF model is well-suited for modeling the matching degree between job

Table 2 Overall performance of GPJFNNF and all baselines for person–job fit in terms of accuracy, precision, recall and F1

Model	Accuracy	Precision	Recall	F1
DT	0.5541	0.5561	0.5458	0.5509
SVM	0.5327	0.5430	0.4268	0.4779
AdaBoost	0.5304	0.5264	0.5224	0.5244
RF	0.5883	0.6049	0.5157	0.5567
GBDT	0.5710	0.5617	0.5956	0.5782
XGBoost	0.5856	0.5778	0.5936	0.5856
CatBoost	0.5850	0.5968	0.5286	0.5606
DSSM	0.7407	0.7121	0.7231	0.7176
Siamese-LSTM	0.7523	0.7542	0.7400	0.7470
ABCNN	0.8341	0.8333	0.8314	0.8323
PJFNN	0.8598	0.8230	0.8821	0.8515
BPJFNN	0.8668	0.8594	0.8462	0.8527
APJFNN	0.8738	0.8524	0.9203	0.8850
SBERT	0.8692	0.8721	0.8721	0.8721
IPJF	0.8691	0.8899	0.8584	0.8739
conSultantBERT	0.8855	0.8971	0.8767	0.8868
InEXIT	0.8528	0.8714	0.8356	0.8531
GPJFNNF (ours)	0.9463	0.9415	0.9504	0.9459

requirements posted by companies and the resumes of job seekers.

Observations were made on the experimental results comparing all classic machine learning methods (i.e., DT, SVM, AB, RF, GBDT). AB shows the weakest result in terms of accuracy while RF performs relatively well. For F1 evaluation metric, SVM shows the weakest result while GBDT performs relatively well. Interestingly, although these the classical classification methods employ the pre-trained word vectors via doc2vec as the input features, their experimental results are still inferior to deep learning-based approaches. This mean that the pre-trained word vectors cannot fully represent the semantic features of the job position texts and the job seeker's resume texts.

Comparing traditional machine learning methods (i.e., DT, SVM, AB, RF, GBDT) with deep learning methods (i.e., DSSM, Siamese-LSTM, ABCNN, PJFNN, etc.), it is observed that the overall performance of deep learning methods is superior to traditional machine learning methods. We believe that the major reason for these results is that the deep models may learn much more efficient semantic features than the shallow models from recruitment data. The experimental results demonstrate that deep neural network methods can capture the semantic features better than traditional machine learning methods in our person–job fit task, which can be beneficial to boosting the final performance.

Among all deep neural network methods, the performance of DSSM is the worst in terms of accuracy, precision, recall,

as well as F1 scores. It is believed that DSSM does not perform well because it cannot capture the sequential texts well in the person–job fit task. Compared with DSSM, Siamese-LSTM obtain better performance. This indicates that a simple LSTM is capable of representing semantic similarity between the job position texts and the job seeker's resume texts.

ABCNN and PJFNN achieve great success than Siamese-LSTM. Among the person–job fit models, we are surprised to find that the recruitment-oriented approach (i.e., PJFNN) show a better performance than ABCNN and Siamese-LSTM. The expresses that the joint representation learning method of the job position and the resume is useful for this particular person–job task.

Among the person–job fit models, PJFNN obtains the worst performance in terms of accuracy, precision, as well as F1 scores. This means that PJFNN model based on CNN may not be a best choice in matching the job position and the resume of job seekers. All in all, to some extent, it is thought that the performance of PJFNN is comparable in terms of accuracy, precision, as well as F1 scores. Compared to PJFNN, BPJFNN further enhances the performance of the person–job fit task. This phenomenon suggests that two BiLSTM are beneficial for learning the semantic representation of each word in job requirements and job seekers' experiences.

It is found that SBERT shows slightly better results in terms of accuracy. In comparison to the BPJFNN model, the SBERT model showed enhancements of 0.24%, 1.27%, 2.59%, and 1.94% in accuracy, precision, recall, and F1, respectively. The main reason for this result may be due to the different encoding methods used in the job position texts and the resume texts. These experimental results indicate that deep learning models based on BERT network perform better than those based on recurrent neural networks.

Further observations were made on the experimental results comparing SBERT with conSultantBERT. In comparison to the SBERT model, the conSultantBERT model exhibited improvements of 1.63%, 2.50%, 0.46%, and 1.47% in terms of accuracy, precision, recall, and F1, respectively. These experimental results indicate that deep learning models based on fine-tuning Siamese Sentence-BERT model perform better than that based on Siamese Sentence-BERT Networks. The findings suggest that fine-tuning Siamese Sentence-BERT model are more suitable for modeling the text information in job requirements and candidate resumes.

The proposed GPJFNNF model was compared with the best-performing method in the person–job fit task, conSultantBERT. The experimental results demonstrate that our GPJFNNF model outperforms the baseline method across four evaluation metrics. The main reason is that the conSultantBERT method fails to effectively learn the global semantic relationships between job requirements and resume texts, as well as the non-Euclidean spatial information.

Table 4 Mappings between the Cliff's δ values and their effective levels

Cliff's delta	Effective levels
$ \delta < 0.147$	Negligible
$0.147 \leq \delta < 0.33$	Small
$0.33 \leq \delta < 0.474$	Medium
$0.474 \leq \delta $	Large

Compared the proposed GPJFNNF model with the InEXIT method in the person-job fit task, the experimental results prove that the GPJFNNF model significantly outperforms the InEXIT method across four evaluation metrics. It is believed that the poor performance of the InEXIT method is due to its lack of consideration of the influence of corporate historical recruitment experience on person-job matching. In our experiments, InEXIT did not perform as well as reported by Shao et al. The reason might be that the InEXIT method is more suitable for handling semi-structured multivariate attribute recruitment data.

It is happy to find that the introduction of graph neural networks is beneficial for specific person-job fit tasks oriented towards talent recruitment. Our model has two main advantages. On the one hand, the GPJFNNF model can effectively learn the semantics of job requirements and the global semantic relationships of applicants' resume texts. On the other hand, the use of the KAN network to fuse the dual-perspective features of local and global semantics of resume texts enhances the semantic representation capabilities of applicants' resumes. The experimental results indicate that exploring the person-job fit task from another novel perspective significantly improves performance.

Ablation study

To demonstrate the effectiveness of each module further, the ablation study of the proposed GPJFNNF model is reported. Two variants of GPJFNNF are designed: (A) GPJFNNF(w/o KAN), removing the KAN network, and (B) GPJFNNF(w/o GNN), referring to removing the GNN network. The ablation study results are shown in Table 3.

Experimental results show that:

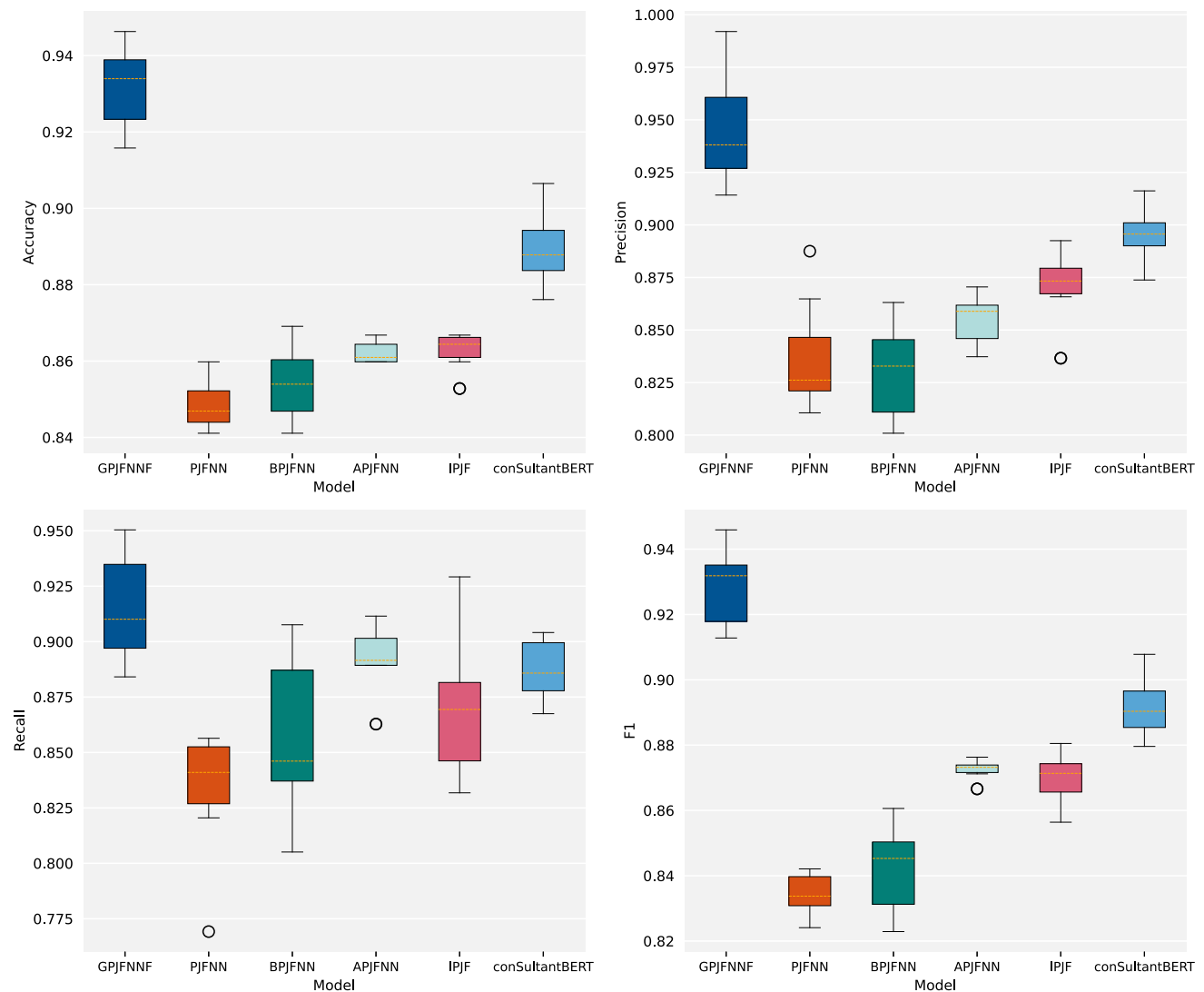
- (1) Regardless of which component is removed, the model's performance decreases, confirming the effectiveness of each component.
- (2) For GPJFNNF (w/o KAN), the accuracy, precision, recall, and F1 score decrease by 2.81%, 2.01%, 3.79%, and 2.90%, respectively indicating that the adaptive feature fusion mechanism effectively integrates both local and global information from resumes.

Table 5 Significance test of GPJFNNF and the top five baselines

	GPJFNNF vs. PJFNN	GPJFNNF vs. BPJFNN	GPJFNNF vs. APJFNN	GPJFNNF vs. IPJF	GPJFNNF vs. conSultantBERT
Accuracy	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)
Precision	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)
Recall	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)
F1	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)	< 0.05 (+Large)

Table 3 Ablation study of GPJFNNF with various modules on person-job fit

Model	Accuracy	Precision	Recall	F1
GPJFNNF	0.9463	0.9415	0.9504	0.9459
GPJFNNF (w/o KAN)	0.9182	0.9214	0.9125	0.9169
GPJFNNF (w/o GNN)	0.8306	0.8248	0.8345	0.8296

**Fig. 2** Boxplot of the four metrics for GPJFNNF and the top five baselines

- (3) For GPJFNNF (w/o GNN), the accuracy, precision, recall, and F1 score decrease by 11.57%, 11.67%, 11.59%, and 11.63%, respectively highlighting the effectiveness of the GNN in capturing global information from resumes.

Statistical analysis

In this paper, repeated experiments will be conducted to compare the proposed model with the top five baselines in terms

of accuracy, precision, recall, and F1. The performance of GPJFNNF, PJFNN, BPJFNN, APJFNN, IPJF, and conSultantBERT will be analyzed by Wilcoxon signed-rank test and Cliff's delta test.

The Wilcoxon signed-rank test is a non-parametric statistical hypothesis test that does not assume data to follow a normal distribution. It is used commonly to compare whether two paired samples come from the same distribution. Compared to the Student's t-test, the Wilcoxon signed-rank test is more sensitive to statistical analysis and more effective in

detecting significant differences. The p -value is utilized to evaluate whether the difference between two paired samples is statistically significant. The p -value < 0.05 indicates a significant difference.

Cliff's delta test [47], a non-parametric effect size measure serves as a complementary analysis to the Wilcoxon signed-rank test in this paper. It quantifies the difference between two populations in numerical terms. Table 4 illustrates the relationship between the Cliff's δ values and their corresponding effect sizes.

Table 5 presents the comparison results of the proposed GPJFNNF model and the top five baselines across four evaluation metrics. The first column represents the evaluation metrics. Columns 2 to 6 display the performance comparison between the proposed GPJFNNF model and other baselines. < 0.05 indicates that the p -value is less than 0.05 in the Wilcoxon signed-rank test, suggesting the results are statistically significant. (+Large) indicates that, under the effective level of Cliff's delta, the GPJFNNF method has achieved a significant performance improvement compared to other methods, with a large margin. The "+" and "-" symbols indicate the direction of the performance difference between the two compared models. A positive value means the performance of the first model is generally better than that of the second model, while a negative value means the opposite. The results for all metrics show that the proposed GPJFNNF model outperforms the other baselines.

Finally, the performance distribution of these top five models across the four metrics is presented using boxplots, as shown in Fig. 2.

Conclusion

Accurate person-job fit has a significant impact on the development of enterprises. Recruiters can identify the most suitable candidates for specific positions, such as software engineers, based on the resumes of job applicants. Given the crucial role that resumes play in companies, it is a significant challenge to automatically and precisely match the right talent to the appropriate job by analyzing the text in job descriptions and applicant resumes.

This paper proposes a Graph-based Person-Job Fit Neural Network Fusion (GPJFNNF) model. It is anticipated that the proposed GPJFNNF model will enhance the online recruitment process. The model consists of three main steps. Specifically, first, the BERT model is used to generate the local semantic representation of job-resume pairs. Second, a graph neural network and attention mechanism are introduced to construct the global semantic representation of resumes. Simultaneously, the KAN network is used to fuse the local and global semantics of the resume. Finally, the fused semantic representation of the resume and the job

description semantics are fed into the person-job fit layer, where the fit result is calculated based on semantic similarity. Extensive experiments were conducted on real recruitment datasets to evaluate the proposed model. Compared to several baselines, the experimental results clearly demonstrate the effectiveness of the GPJFNNF model in terms of accuracy, precision, recall, and F1 scores.

In the future, it is planned to explore the dynamic interaction of multi-attribute variables between job requirements descriptions and job seekers' resumes to capture semantic associations between job postings and resumes. Additionally, the person-job fit task will focus on developing more competitive text representation methods to enhance the performance. Furthermore, it is intended to extend the solutions of the person-job fit task to other text matching tasks.

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Data Availability The data used to support the findings of this study are included within the article. The data used to support the findings of this study are available from the corresponding author upon request.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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